

Visualization Critique and Redesign

From audience to visual form

Information Visualization · Lecture 10

Understanding → Task Fit → Encoding

Design and critique follow three stages

Audience and purpose → Design direction → Visual encoding

1. Understanding

Profile the audience and clarify the task.

Design: Define the reader's background knowledge and what they need to understand.

Critique: Infer the intended reader and task from the title, context and explanations.

2. Task Fit

Match the design style to the reader and task.

Design: Use the Visualization Wheel to choose a suitable balance.

Critique: Mark the existing design's tendencies on the Wheel and judge their fit.

3. Encoding

Choose and evaluate how the graphic represents data.

Design: Select marks and channels, with pictograms and consistent rules where useful.

Critique: Evaluate expressiveness, effectiveness, separability and capacity.

After redesign, revisit the same three stages to assess the result.

Understanding starts with readers and design purpose

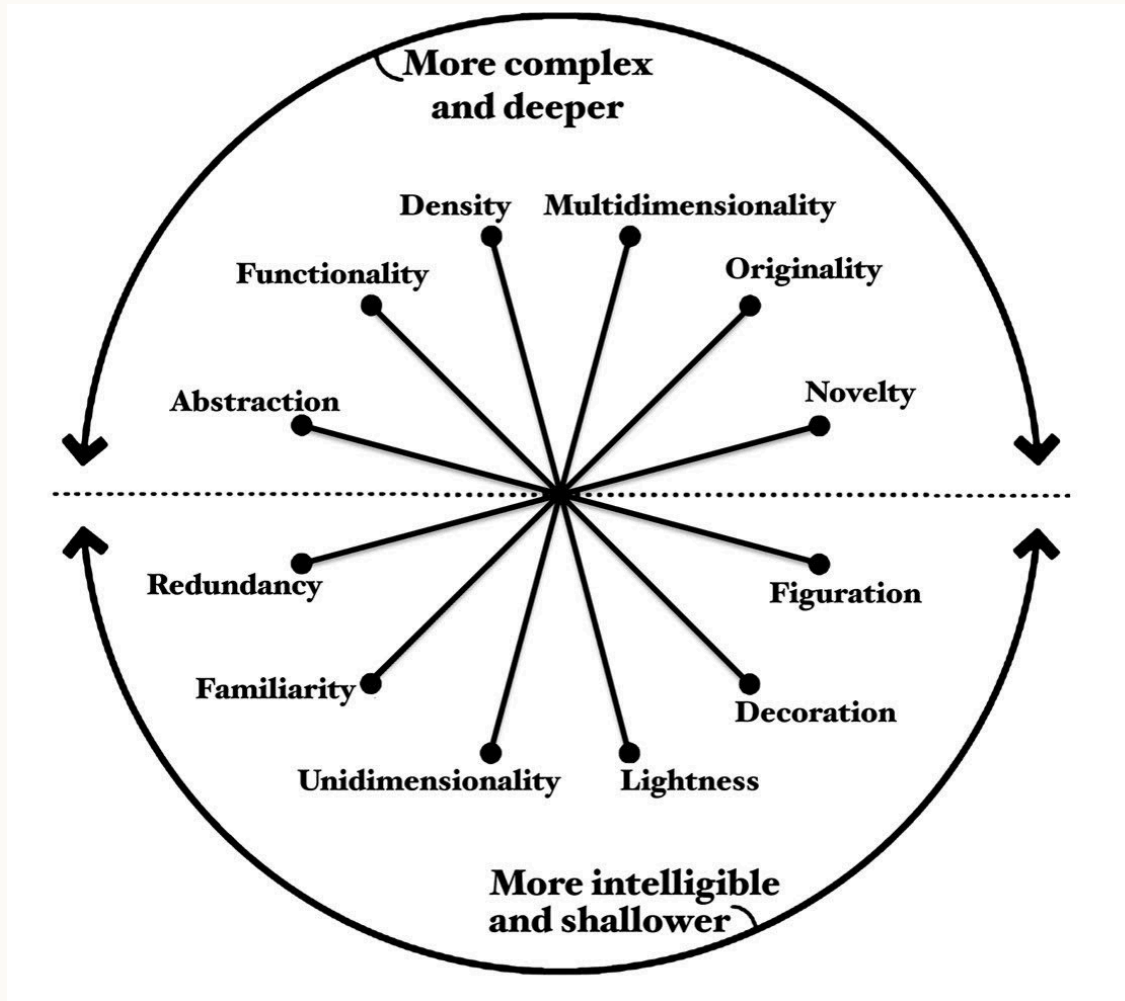
Profile the audience

- **Domain knowledge:** Which concepts and terms do readers already understand? What needs explanation?
- **Graphical knowledge:** Which chart types, scales and visual conventions are familiar to them? What reading rules need an introduction?

Understand the design purpose

- **Purpose:** What should readers learn or do? Identify a trend, compare values, explore patterns or make a decision?
- **Context:** Where and how will they encounter the visualization? Consider the medium, available time and whether someone will explain it.

Task Fit connects design tendencies to the reading task

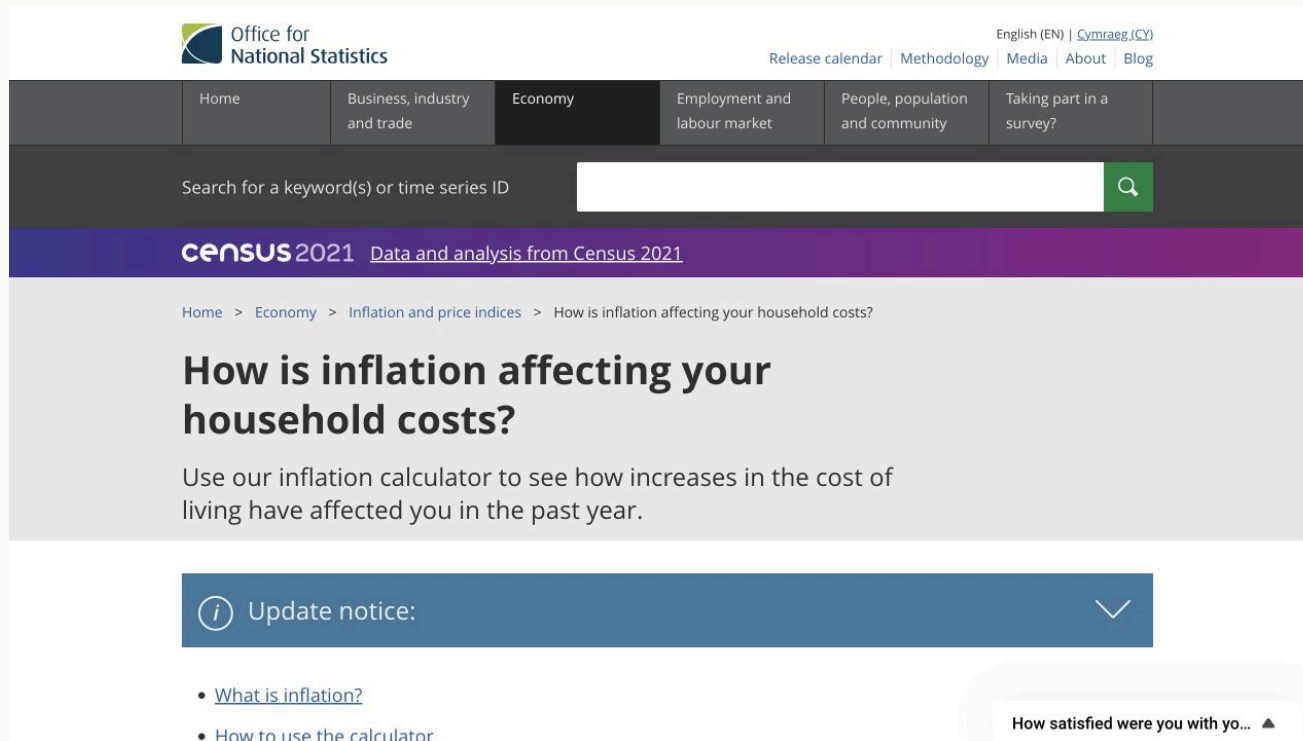


Task fit is the match between a design's choices and what its intended reader needs to understand or do.

The **Visualization Wheel** describes design tendencies. Mark the choices visible in a graphic, then explain whether they help the task.

In redesign, use the same dimensions to state a target direction.

Example: Inflation affects your household cost (ONS)



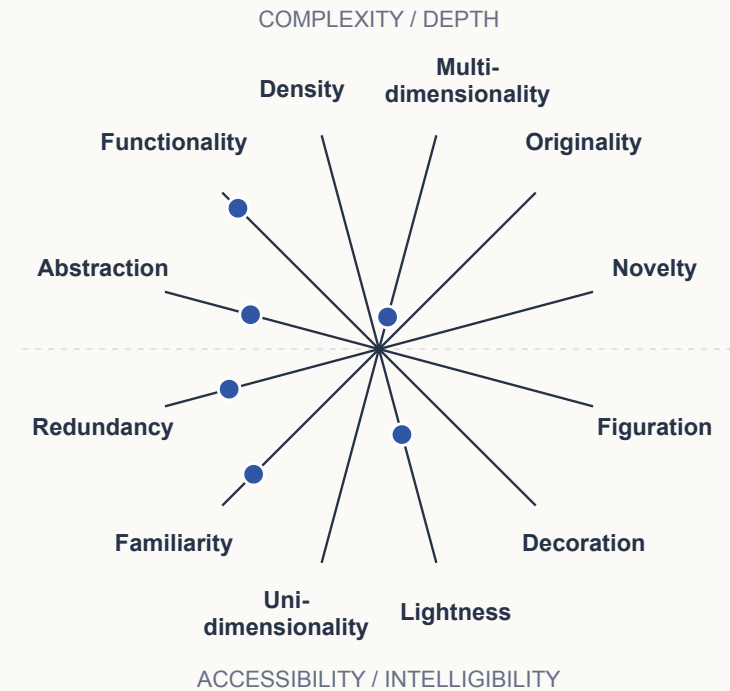
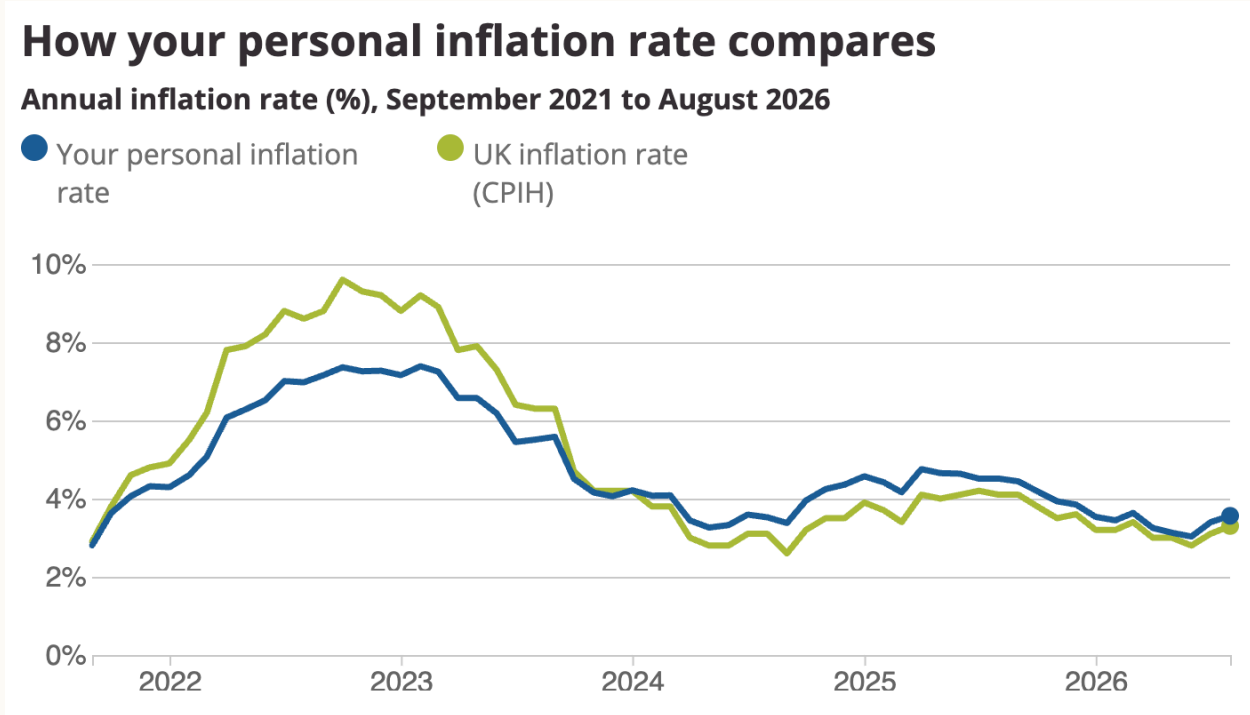
Design purpose

Help readers understand how inflation affects their own household costs using everyday spending information.

Audience

- **Domain knowledge:** Readers have diverse backgrounds and varying knowledge of economics.
- **Graphical knowledge:** Readers have different levels of experience with charts, so use simple, familiar graphics.

ONS combines an abstract chart with familiar guidance

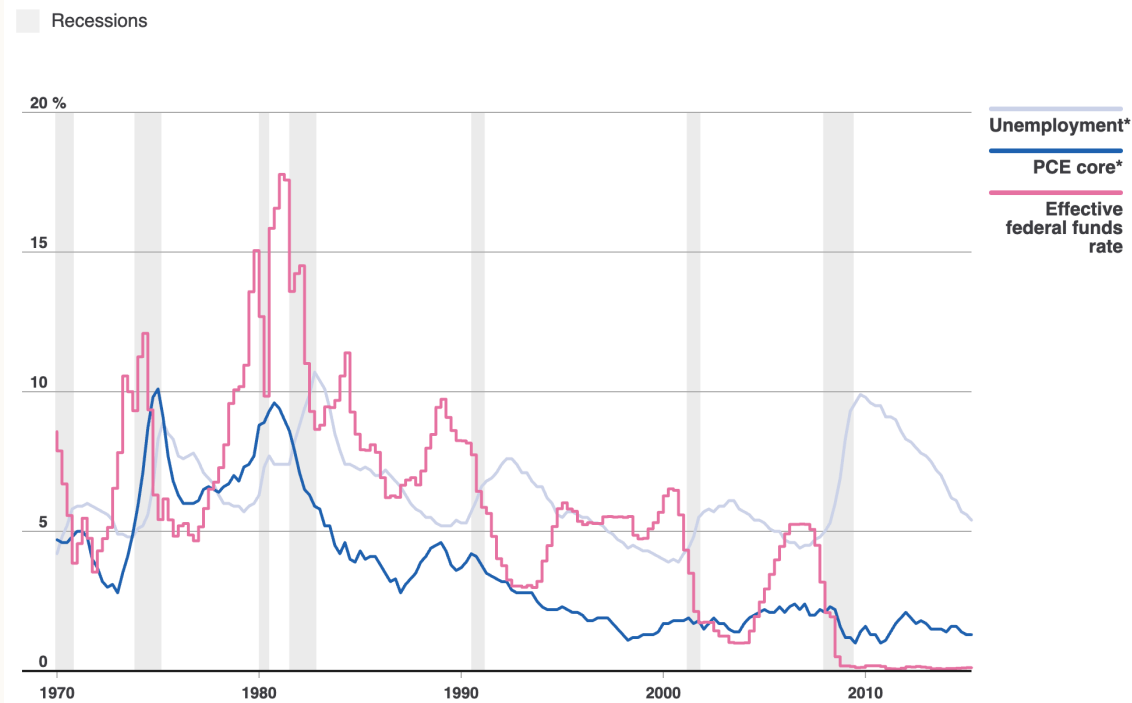


- **Abstraction + familiarity:** Conventional lines compare a personal rate with a UK benchmark. Explain the inflation measure clearly.
- **Lightness + redundancy:** Step-by-step guidance, headings and labels repeat the context and support personal comparison.

Example: Yellen's bet on inflation (Reuters)

Yellen's bet on inflation

The Fed may begin raising the target interest rate as soon as September. But if inflation doesn't also rise, the U.S. economy is at risk of creating a positive "real" interest rate before it's ready.



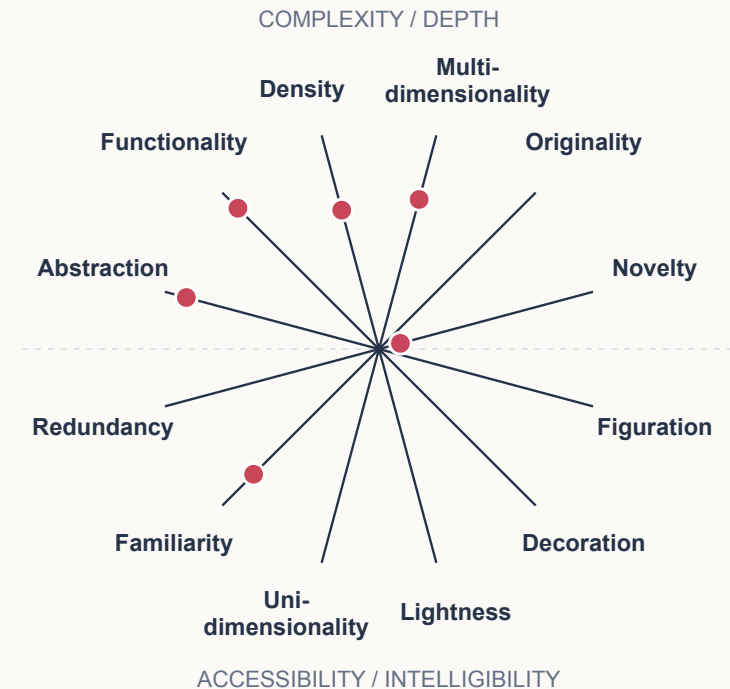
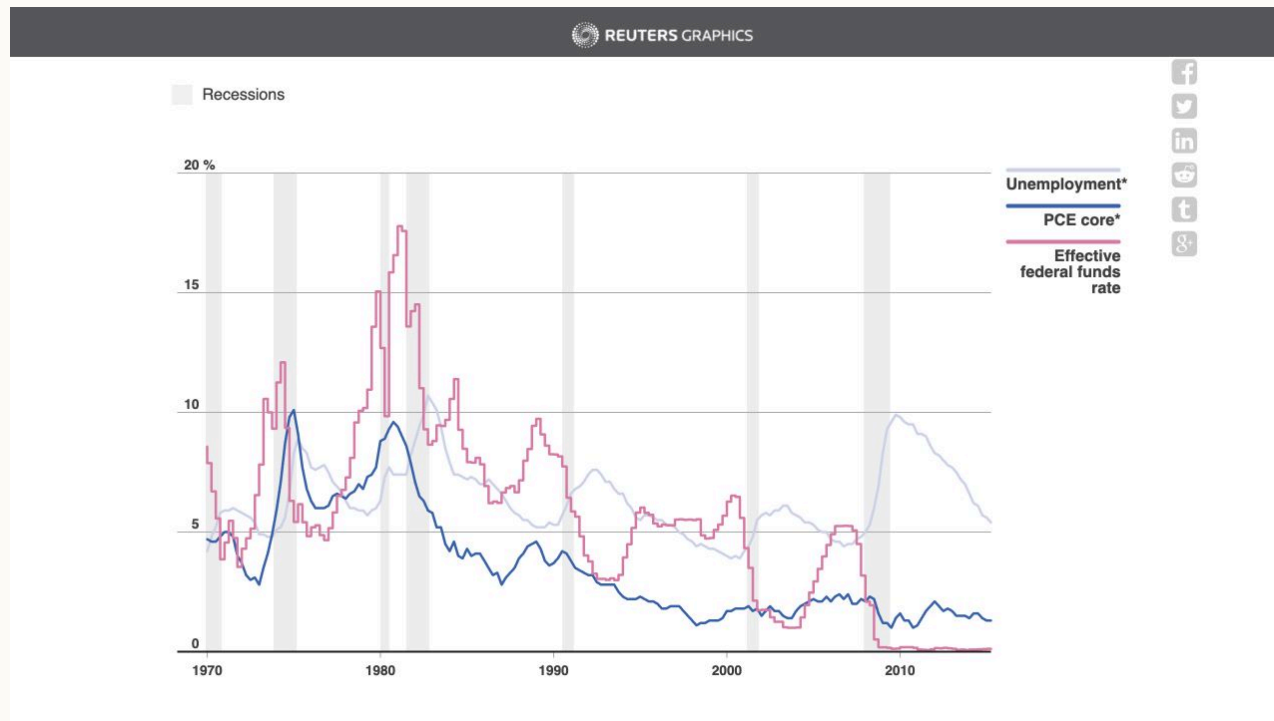
Design purpose

Explain how inflation and interest rates relate to the Fed's decision about raising rates.

Audience

- **Domain knowledge:** Readers are expected to understand economic terms such as PCE core and the federal funds rate.
- **Graphical knowledge:** Because this is a Reuters news report, we should not assume that all readers have strong graphical knowledge.

Reuters pairs greater density with a familiar chart form



- **Density + multidimensionality:** Several measures and recession periods share one timeline, supporting comparison across measures.
- **Abstraction + familiarity:** Conventional lines show technical variables. Readers may still need definitions and a clear reading focus.

Reuters Graphics · Yellen's bet on inflation · 2015 historical example · Data: Federal Reserve Bank of St. Louis · Alberto Cairo · The Functional Art · Visualization Wheel

Encoding: marks and channels

Marks represent items or relationships

Marks are the graphical elements used to represent data, such as points, lines and areas.

- A point can represent a country.
- A line can connect observations over time.
- An area can represent a geographic region.

Channels encode attributes

Visual channels are properties of marks that vary with data, such as position, length, area, color and shape.

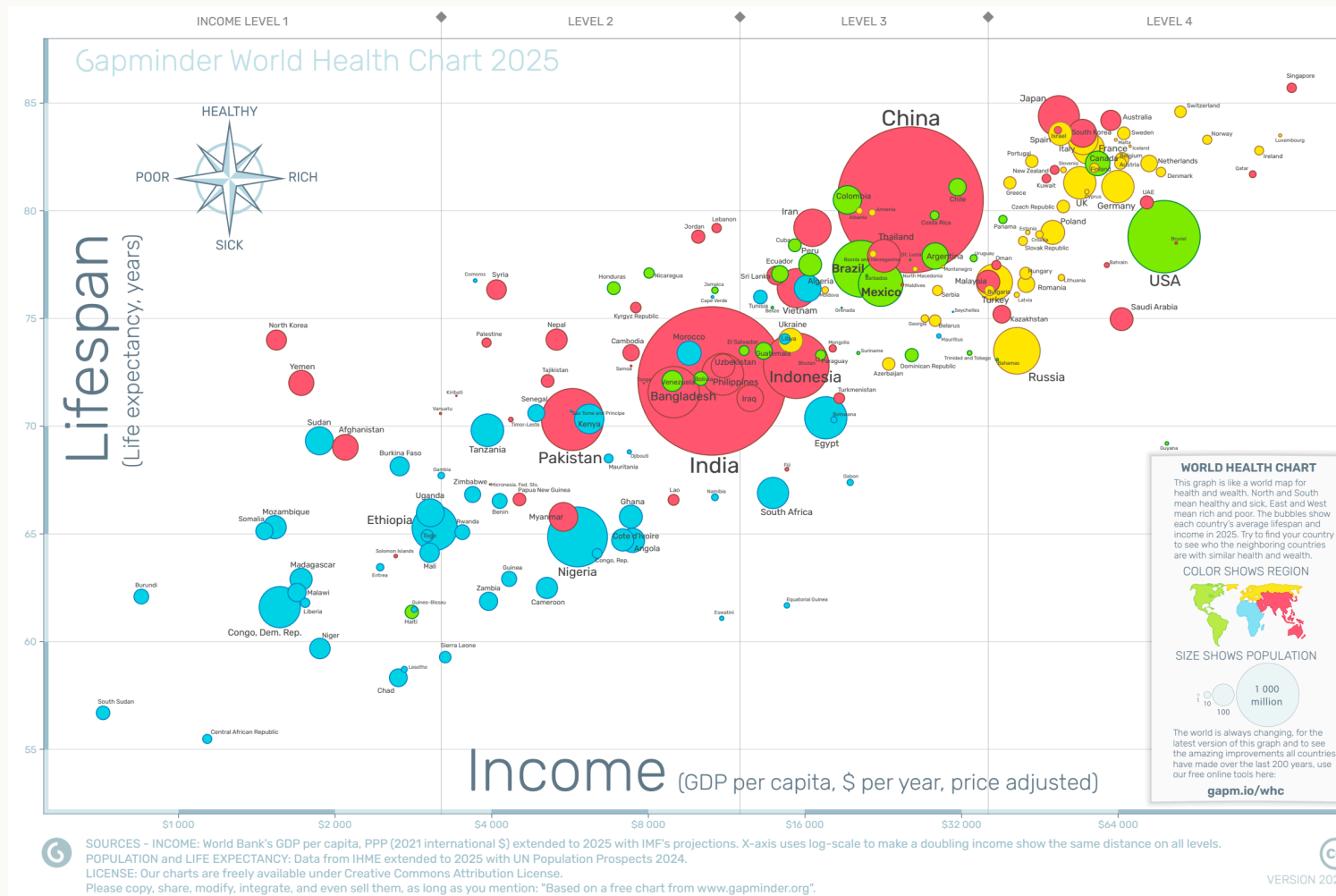
- **Quantitative values:** position or length can show magnitude.
- **Categories:** hue or shape can distinguish identity.

First decode the mapping: What does each mark represent? Which attribute controls each changing property?

Four checks for a visual encoding

Principle	Meaning	Critique question
Expressiveness	Visual relationships must match the relevant relationships in the data.	Does the encoding omit meaning or imply an order or magnitude the data do not have?
Effectiveness	Task-important attributes should use channels that support accurate, efficient judgments.	Does the key comparison receive a strong, suitable channel?
Separability	Viewers can read one channel without substantial interference from another.	Does variation in size make hue or shape harder to read?
Capacity	Capacity is the number of distinct values readers can reliably distinguish through a channel.	Are the differences visible at this size, density and viewing distance?

Gapminder: match each channel to its job



Expressiveness

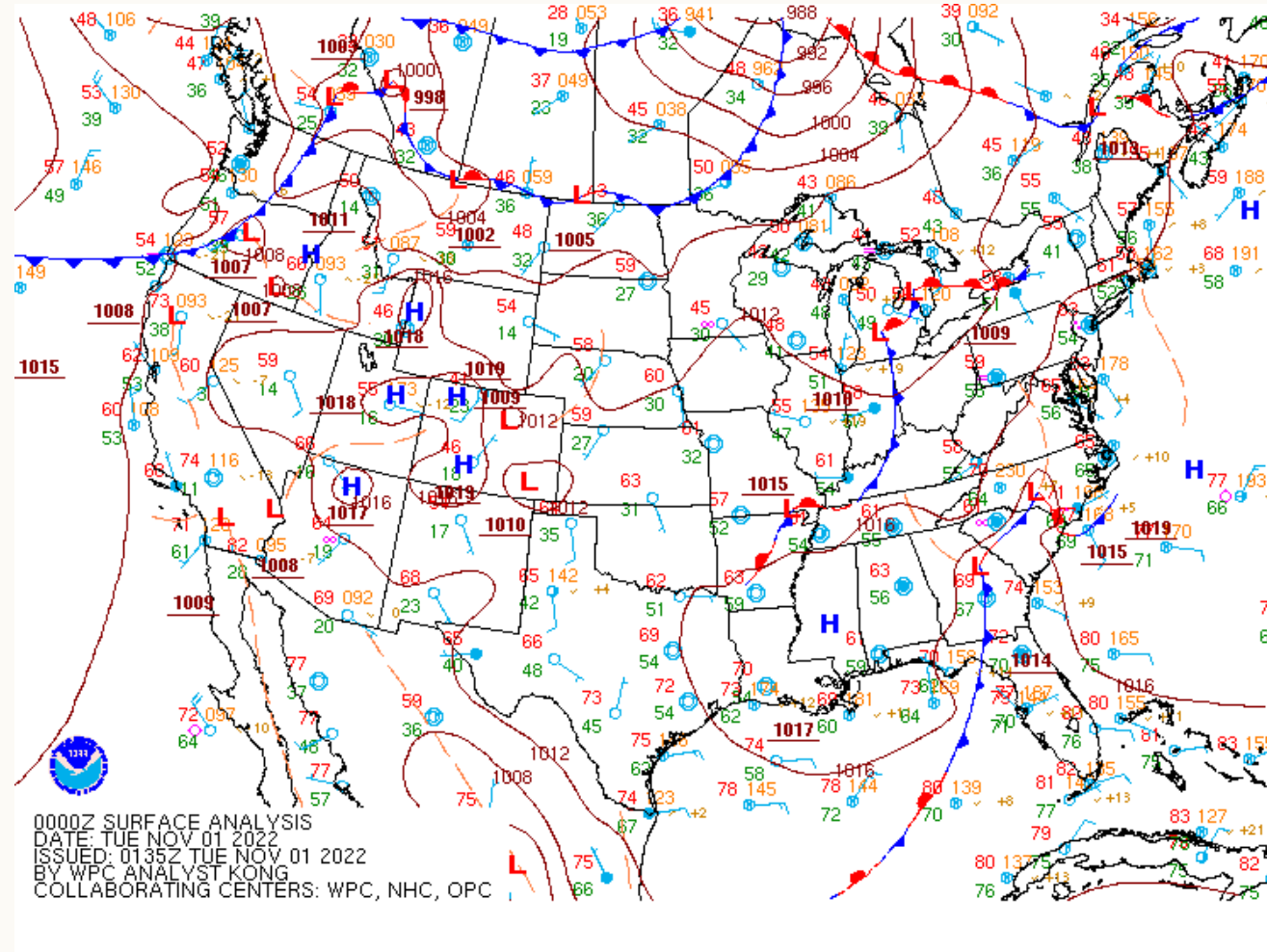
Position shows income and lifespan; hue identifies regions; bubble area represents population.

The income axis uses **equal spacing for equal ratios**, not equal dollar differences.

Effectiveness

Position serves the income–lifespan comparison. If precise population ranking is the task, use a bar or dot chart.

Weather symbols rely on learned conventions



What must the reader recognize?

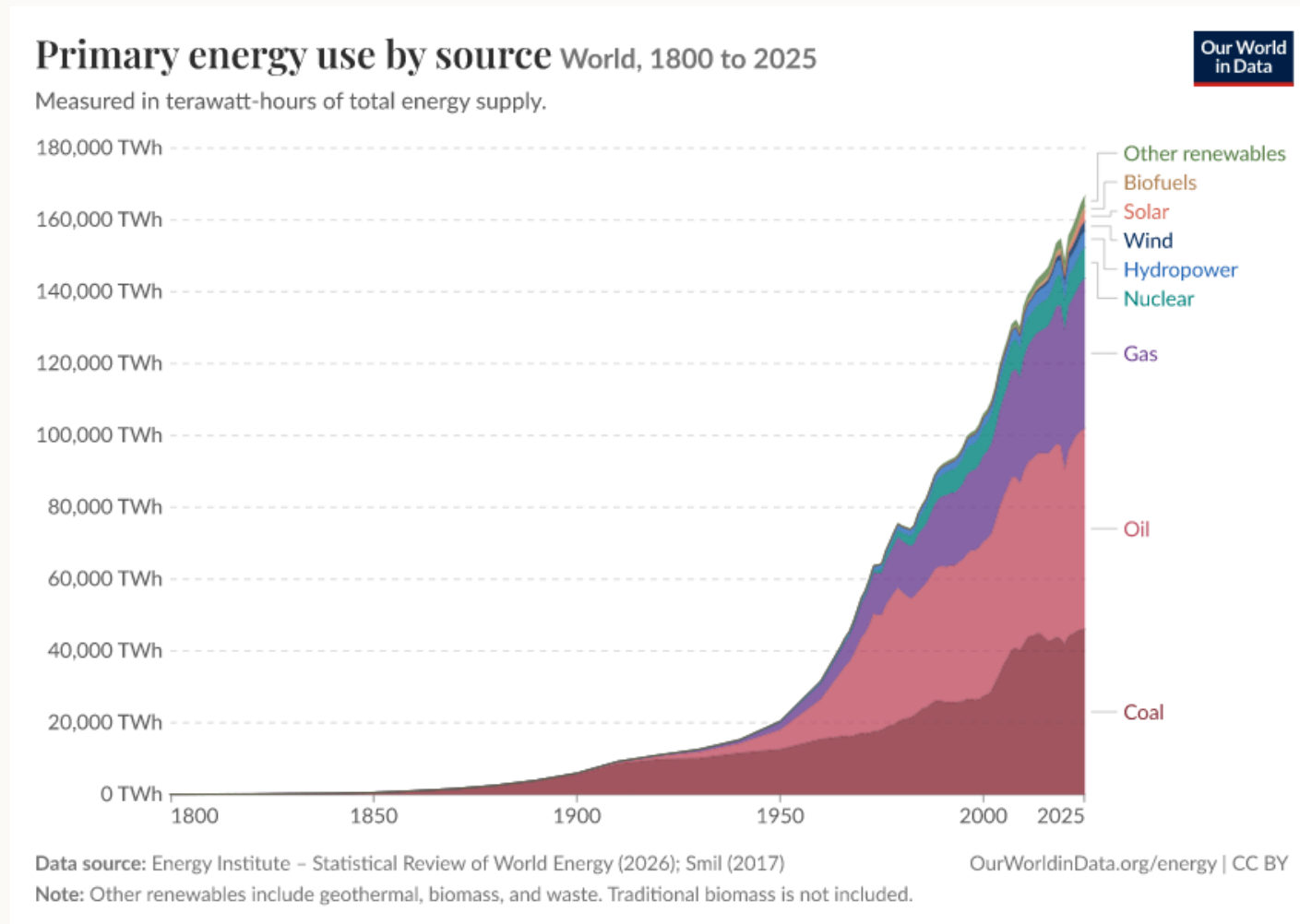
Blue triangles identify a **cold front**. Wind barbs encode direction and speed through a different convention.

Critique → redesign

These symbols can be familiar to a trained weather reader. A general reader may see the shapes without knowing their meaning.

Add a nearby key and explain one symbol; retain standard symbols for specialist use.

Thin bands make hue harder to distinguish



Capacity

The nine labelled colors are within a reasonable range for this chart.

Separability: hue and size

Thin bands make hue harder to distinguish: **size interferes with reading color.**

Redesign

Retain direct labels. Enlarge the small sources in a separate panel or small multiples.

Visualizing electricity sources of European countries

Design task: Show where selected European countries are and how their electricity sources differ.

The data

- **Records:** eight countries in 2024.
- **Attributes:** nine sources of electricity generation, measured in TWh.
- **Derived measure:** each source's share of national generation.

The scope

Denmark, France, Germany, Italy, Norway, Poland, Spain and Sweden.

The comparison concerns **electricity generated domestically**.

The preceding OWID example shows **primary energy**; this case uses a different measure.

The audience profile sets the reading requirements

Primary audience: general readers

- **Domain knowledge:** may know wind and solar, but need terms such as bioenergy explained.
- **Graphical knowledge:** varied; familiar maps, bars and pies still need a clear legend.
- **Purpose and context:** a news or teaching graphic for locating countries and recognizing their main sources.

Secondary use: closer comparison

- Energy analysts may want to compare a particular source or inspect small differences.
- A compact map cannot make every small value easy to read.
- Keep a common unit and provide a companion view with direct values.

Design requirements: country labels, explained categories, a percentage rule and access to numerical detail.

OWID already supports several ways to explore the data

Existing views

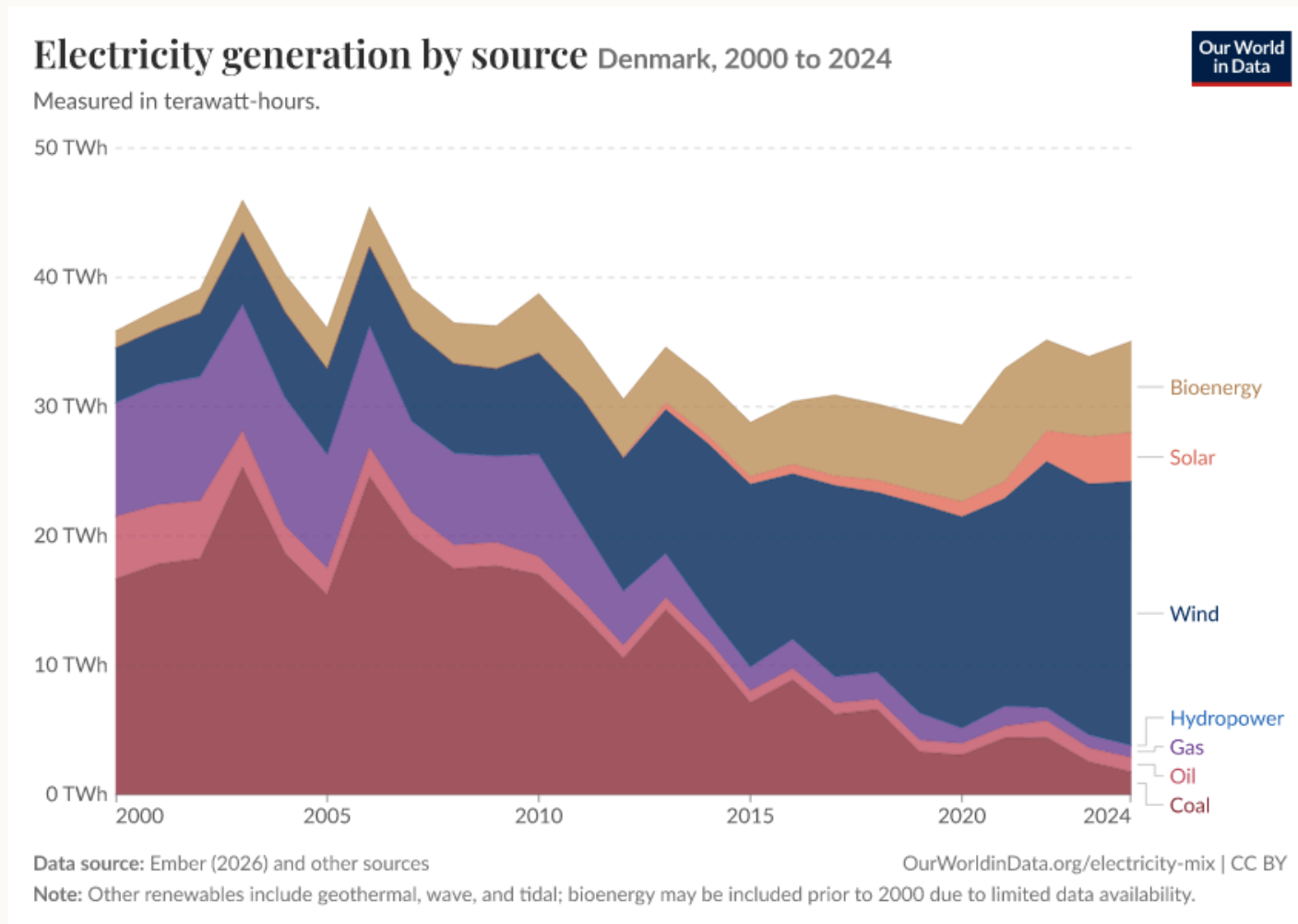
Area charts show generation over time. Share charts compare proportions. Maps show geographic variation in a selected source.

This chart

One country, several sources, many years. Axes and direct labels explain how to read it.

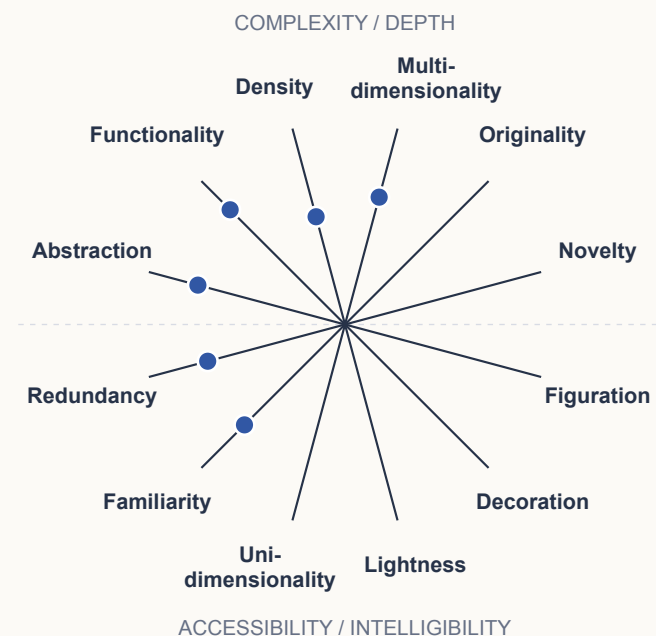
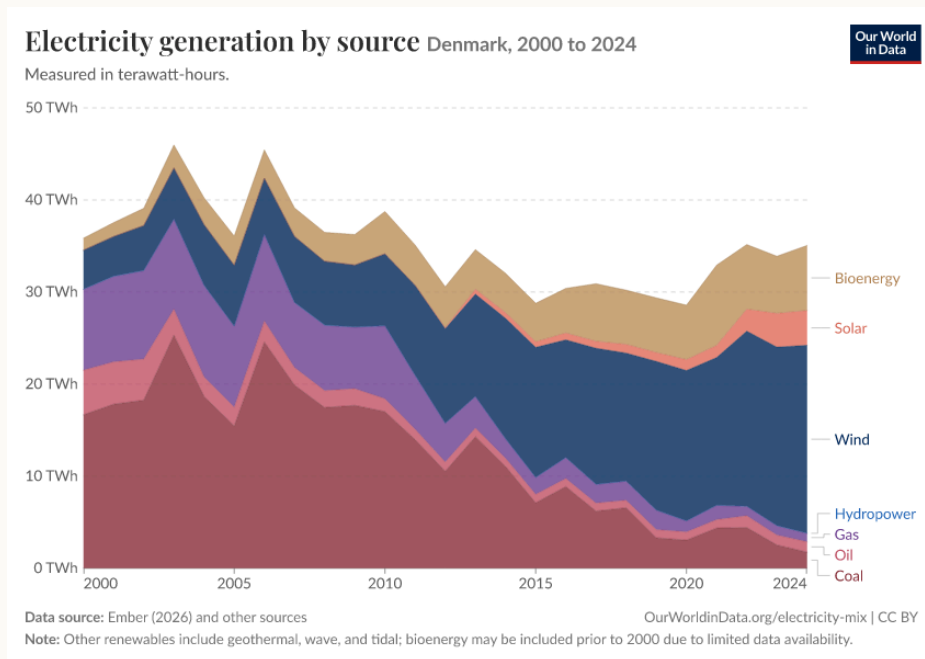
Our question

Where are the selected countries, and how do their full mixes compare in 2024?



OWID · Electricity generation by source · Denmark, 2000–2024 · Ember (2026) and other sources · CC BY

The existing style is functional, familiar and information-dense



- **Functionality + familiarity:** conventional axes and direct labels help readers interpret the data. Keep these strengths.
- **Density + multidimensionality:** many years and sources serve trend exploration; our brief needs country location and same-year composition instead.

OWID · Electricity generation by source · Denmark, 2000–2024 · Ember (2026) and other sources · CC BY

Visualization Wheel: Alberto Cairo · Dots show a qualitative design assessment

The redesign places country profiles in geographic context

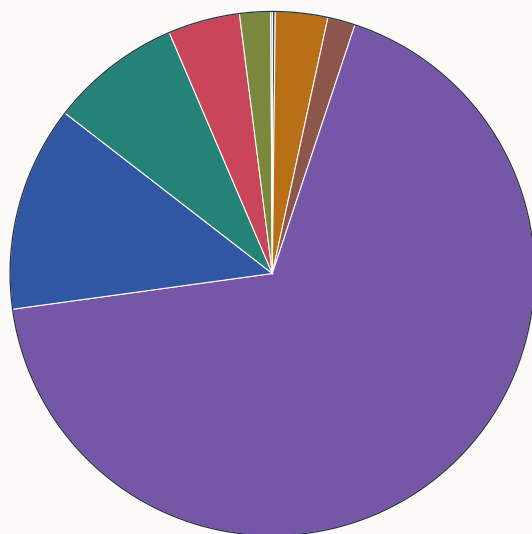
Requirement from the brief	Design response
Locate the selected countries.	Use a map as the arrangement for the country profiles.
Recognize the full source mix.	Place one compact, multivariate glyph at each country.
Support readers with varied knowledge.	Keep familiar charts, country names and a shared source legend.
Compare a source more precisely.	Add a separate view with aligned values and direct labels.

Next decision: should each country use a pie glyph or a bar glyph?

Data: [Ember \(2026\)](#), processed by [Our World in Data](#) · 2024 · Share of domestic electricity generation

One country's nine shares can form a pie or bar glyph

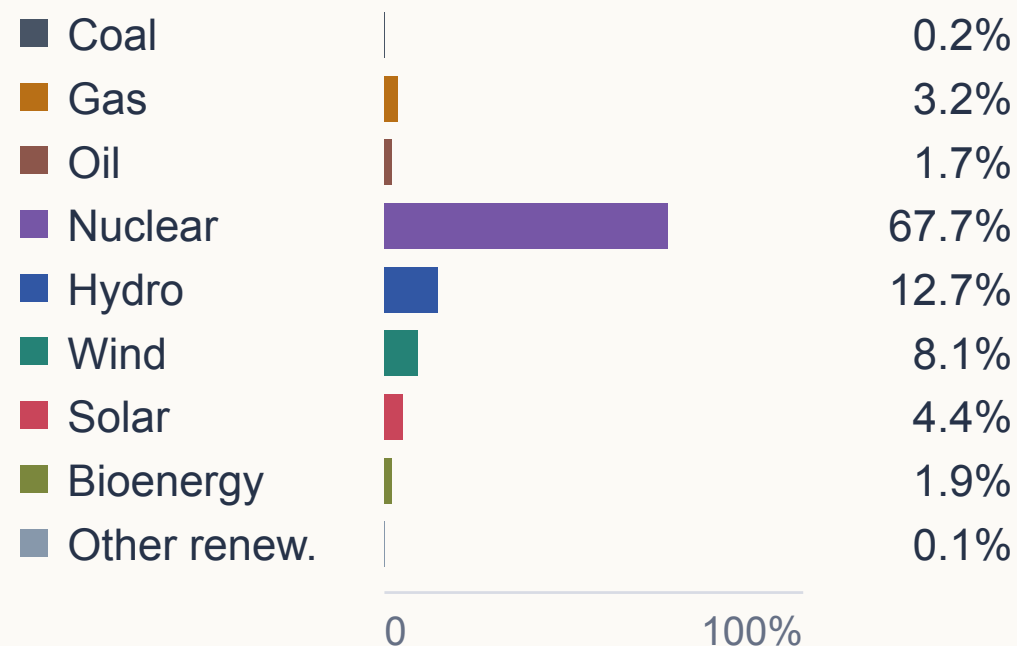
France · pie glyph



Sector angle and area = share

The whole circle represents 100%.

France · bar glyph

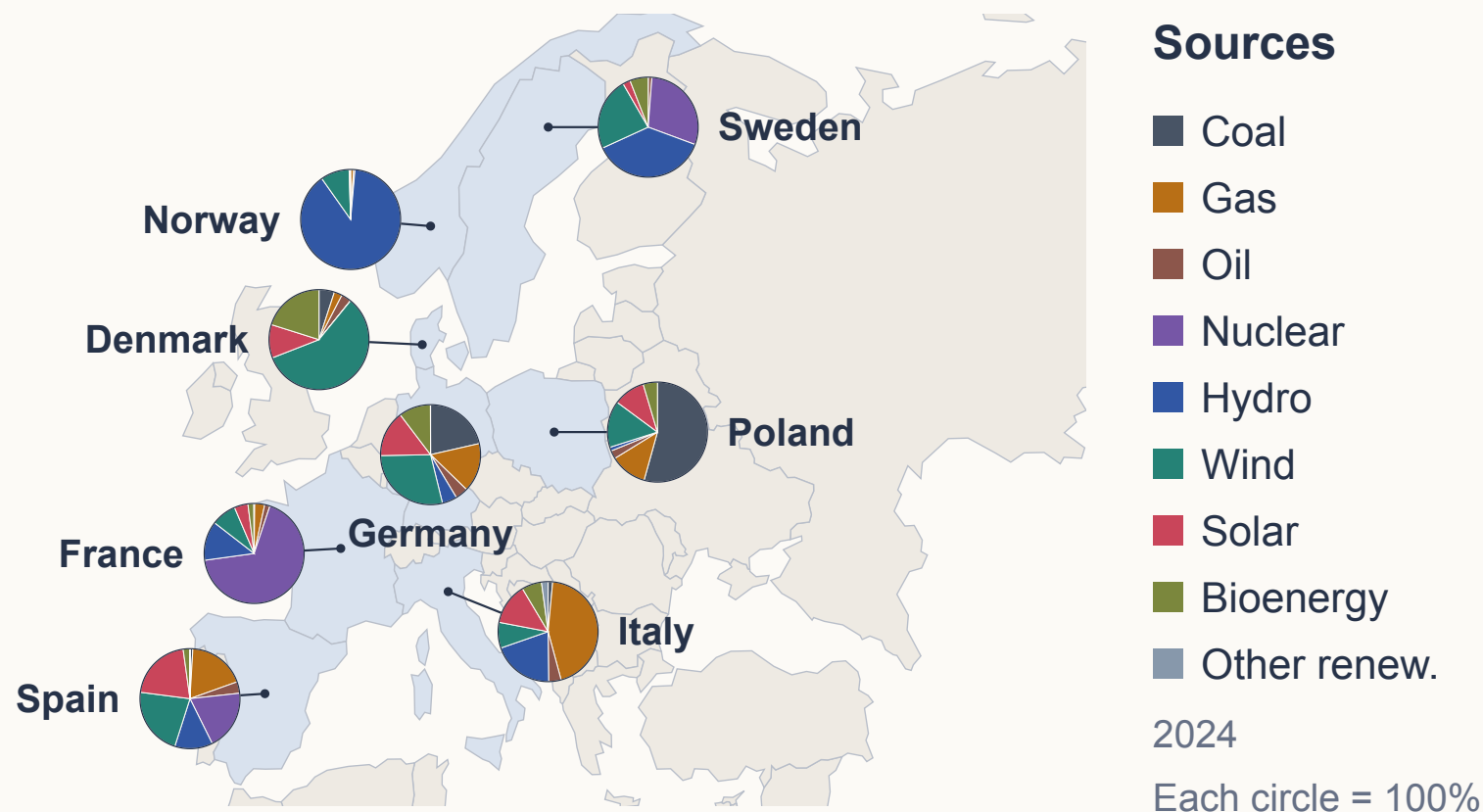


Length = share · a common zero baseline

Country position will locate the record; color identifies sources; sector size or bar length represents share.

Data: [Ember \(2026\)](#), processed by Our World in Data · 2024 · Share of domestic electricity generation

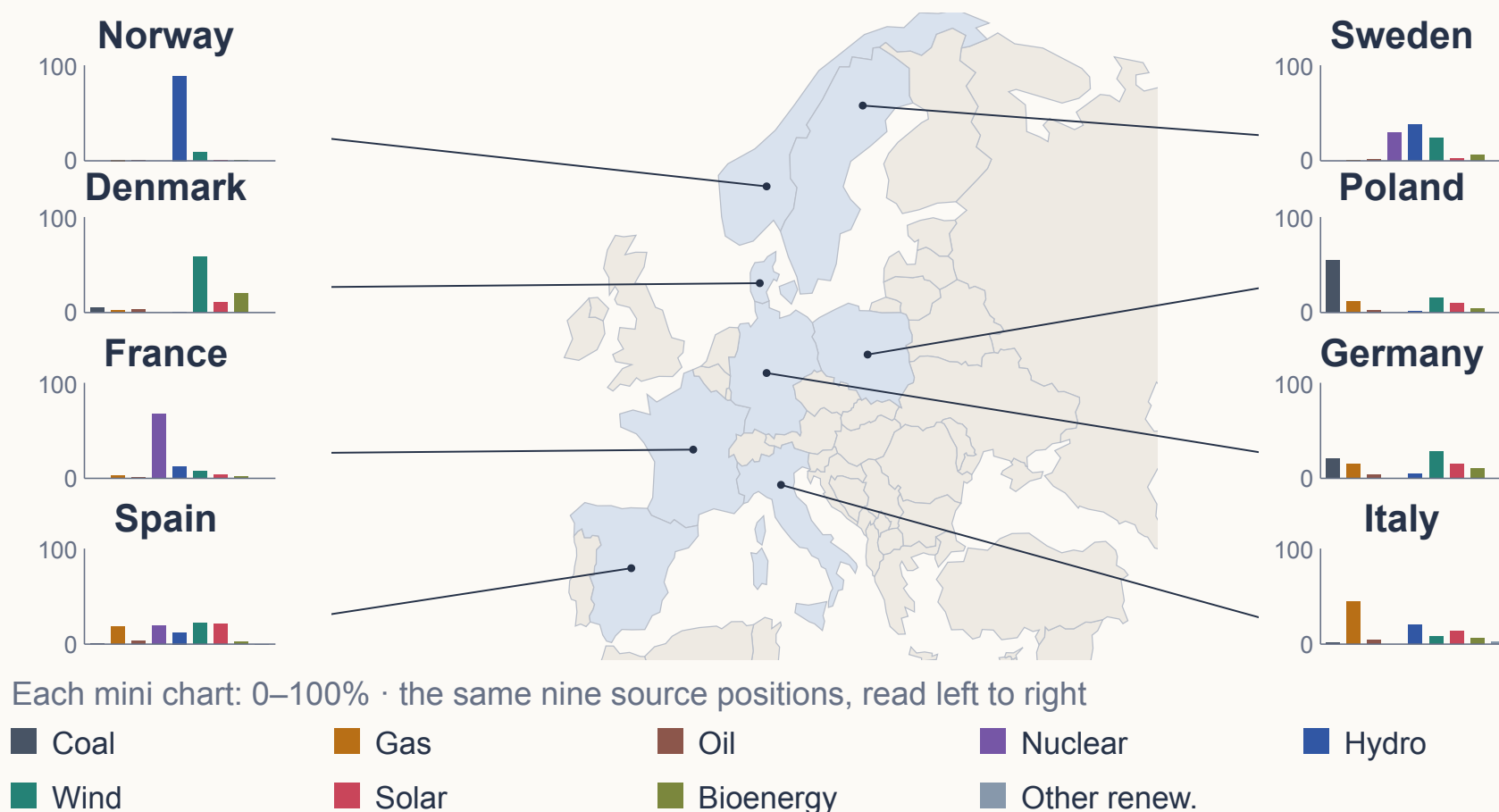
Pie glyphs show the whole mix in a compact form



- Part-to-whole: recognize dominant sources and the complete mix.
- Compact form: show country profiles together in geographic context.

Data: Ember (2026), processed by OWID · 2024 · Map: Natural Earth

Bar glyphs support comparison on a common scale

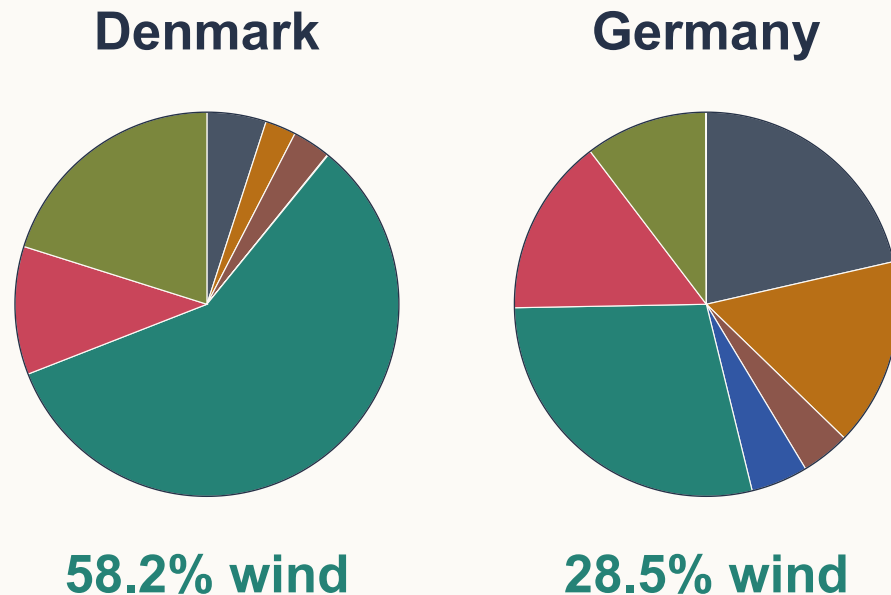


Useful task: compare a source's share across countries, or compare sources within one country.

Data: Ember (2026), processed by OWID · 2024 · Map: Natural Earth

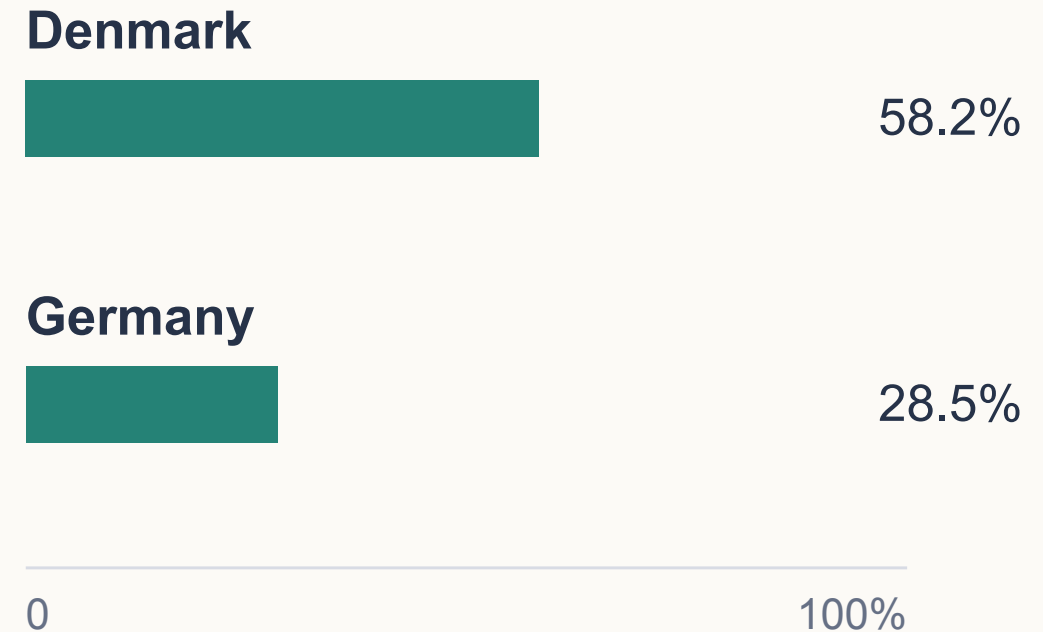
The task determines which quantitative comparison matters

Wind inside the full composition



Pies retain the part-to-whole context.

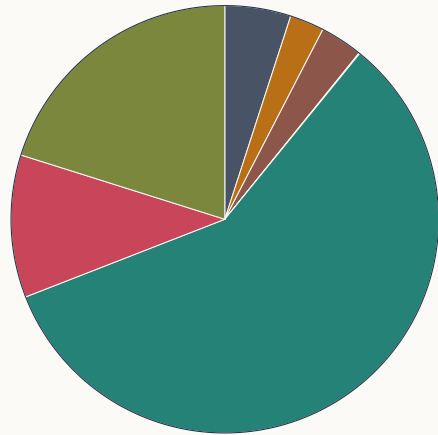
Wind on a shared baseline



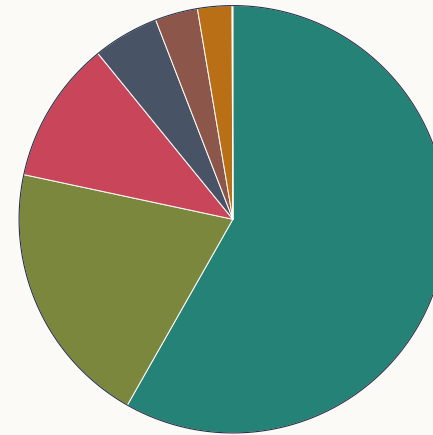
Aligned lengths make the comparison direct.

Stable category positions make glyphs easier to compare

Fixed order across countries



Sorted separately in each country



Same category sequence and start angle.

The same source moves to a new position.

Both circles show Denmark's same values. Keep the mapping stable for comparison.

Coal

Gas

Oil

Nuclear

Hydro

Wind

Solar

Bioenergy

Other renew.

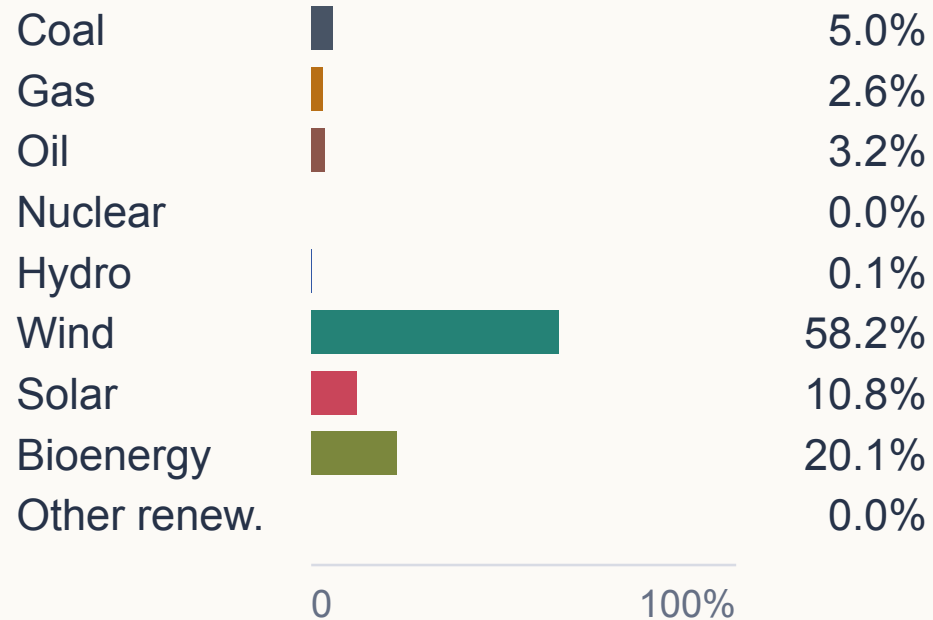
Keep the year, denominator, glyph size, colors and quantitative scale consistent as well.

Data: [Ember \(2026\)](#), processed by [Our World in Data](#) · 2024 · Share of domestic electricity generation

Stable category positions make bar glyphs easier to compare

Fixed order bar glyph

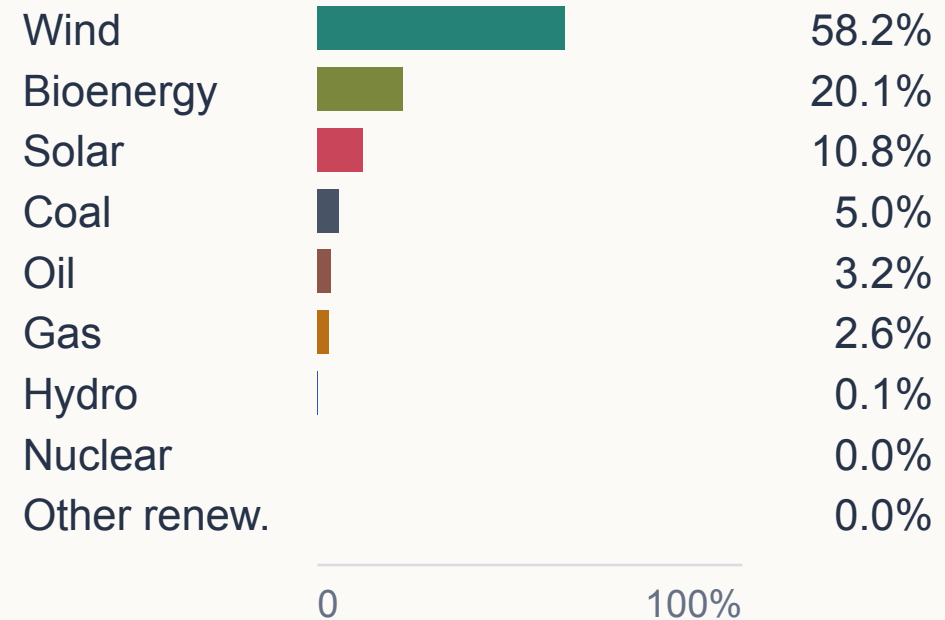
Denmark · 2024



Match a source across country glyphs.

Sorted bar glyph

Denmark · 2024

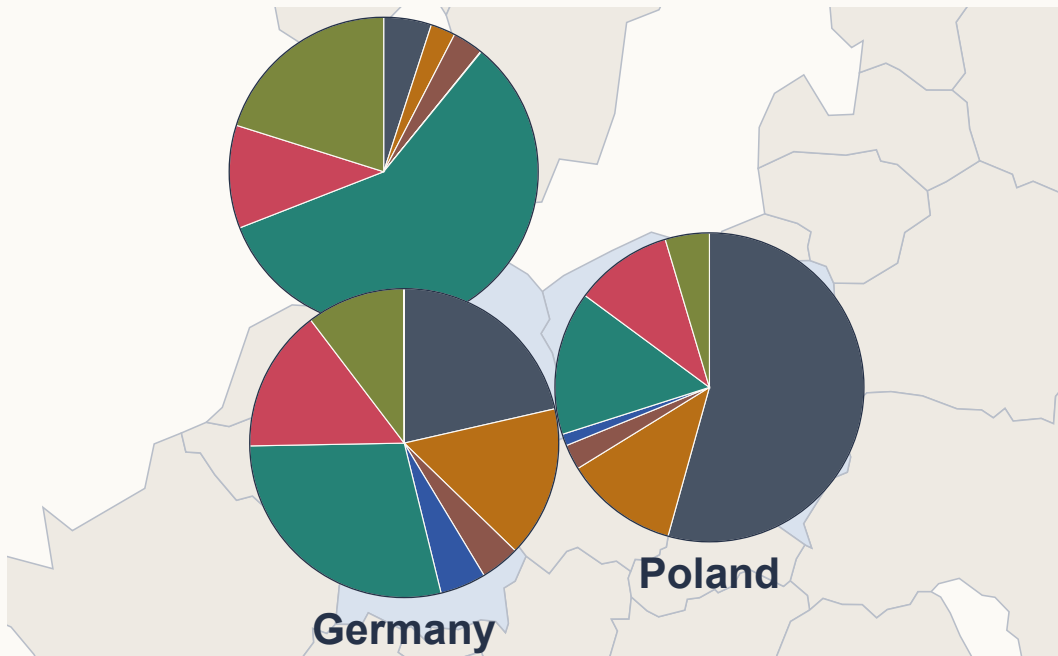


Read the ranking within one country.

Same Denmark data · same 0–100% scale · only the row order changes.

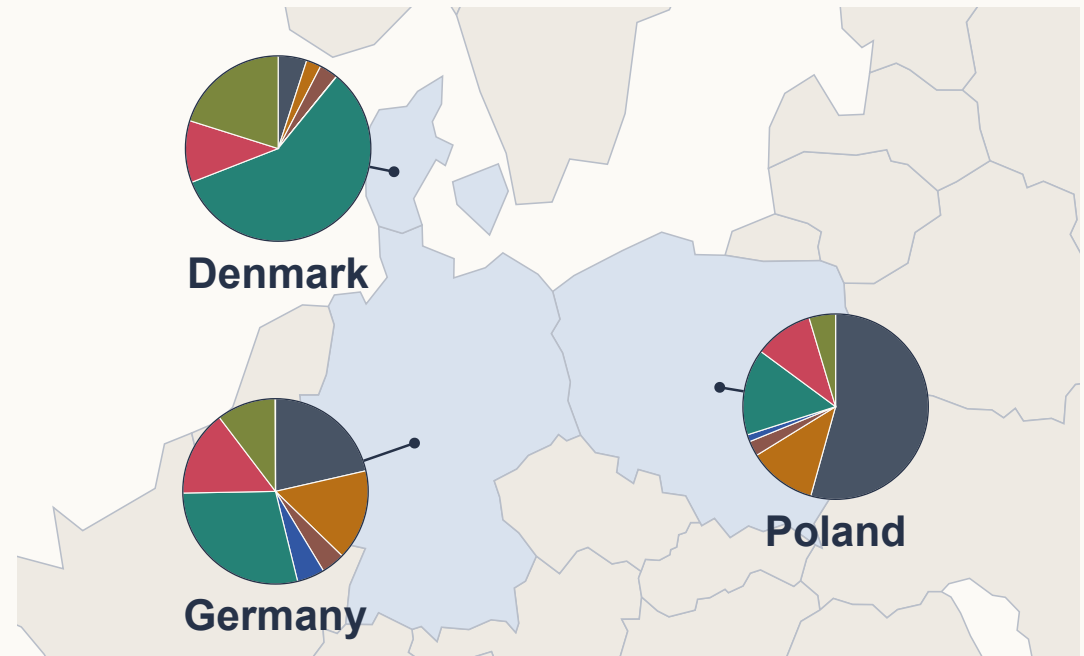
Glyph size and placement must preserve readable marks

Large glyphs at country anchors



Occlusion hides slices and country boundaries.

Smaller glyphs with clear links



Move the glyph; keep its country anchor.

Pictograms can make the visualization more engaging



■ Coal



■ Gas



■ Oil



■ Nuclear



■ Hydro



■ Wind



■ Solar



■ Bioenergy



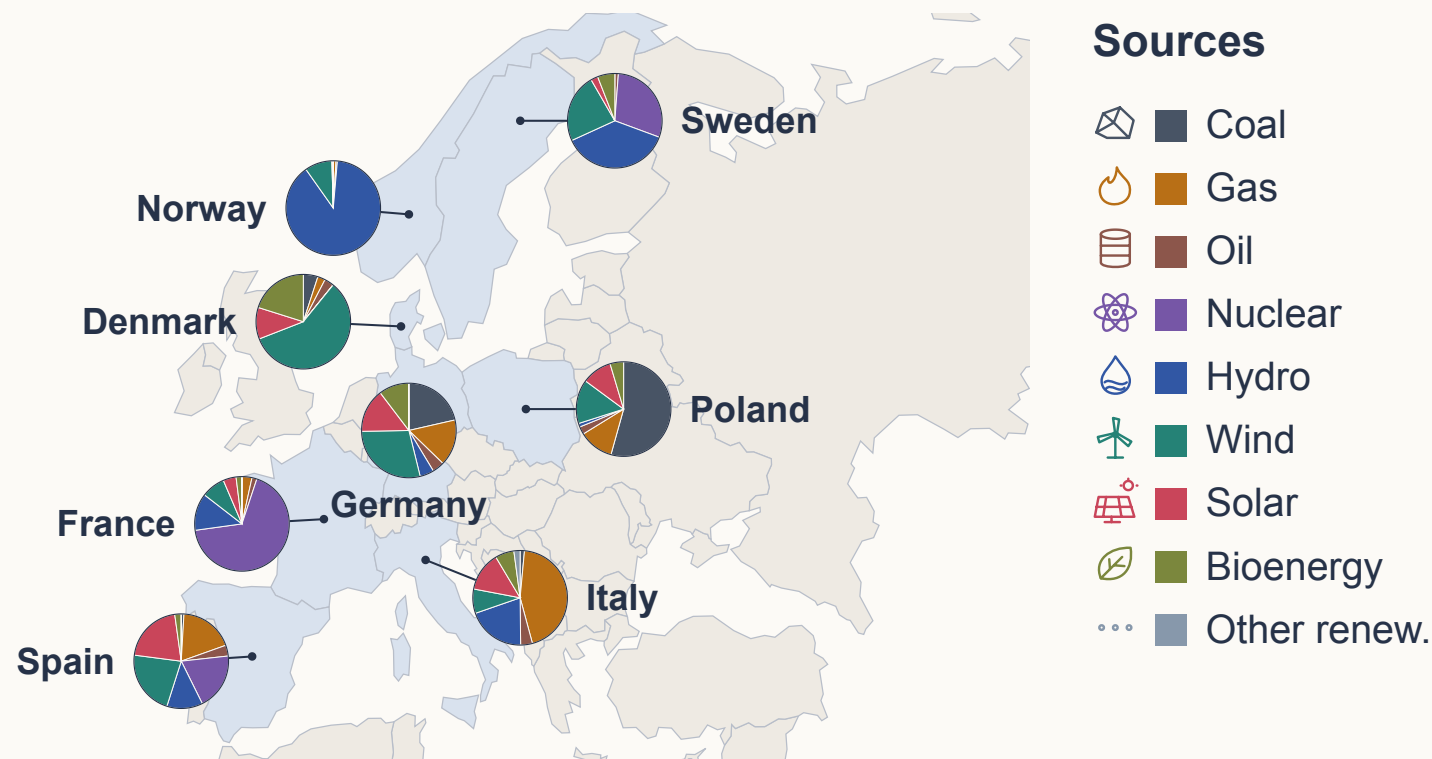
■ Other renew.

Pictogram + name + color identify a source. The glyph's bars or sectors encode its share.

Here, pictograms act as **ornament**: decorative elements that add visual character.

Data categories: [Ember \(2026\)](#), processed by Our World in Data · Electricity sources

A consistent glyph map supports geographic comparison



2024 · selected countries · each circle = 100% of national electricity generation

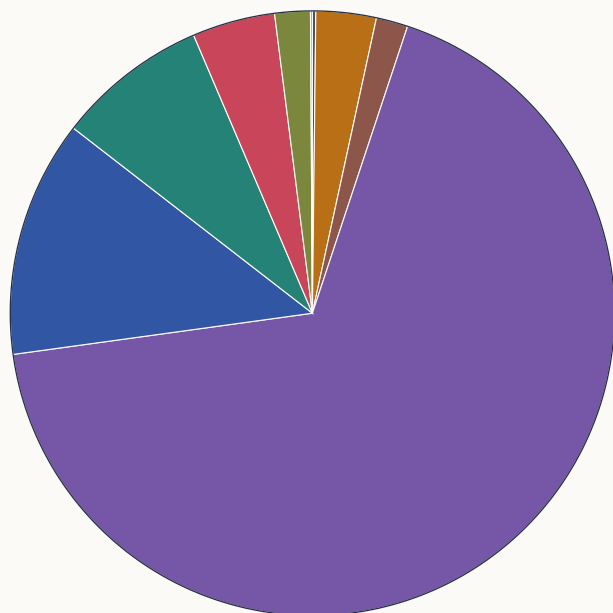
Equal circle size · fixed source order · background countries carry no values

Solid color swatches make hue easier to distinguish than pictogram outlines alone.

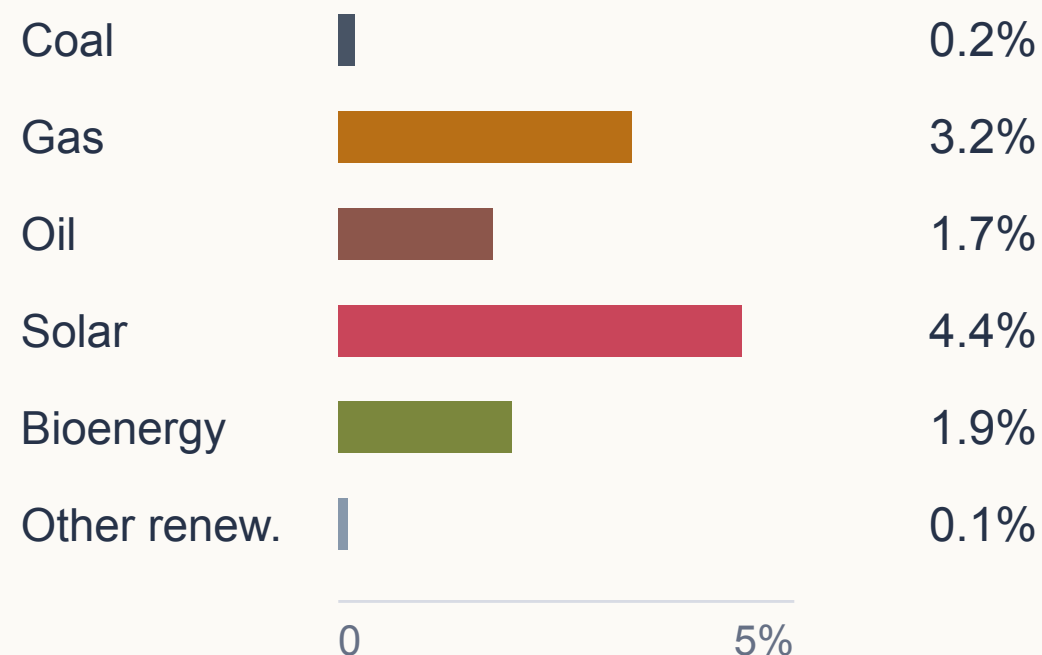
Data: Ember (2026), processed by OWID · 2024 · Map: Natural Earth

Small slices need labels or a detail view

France · complete mix



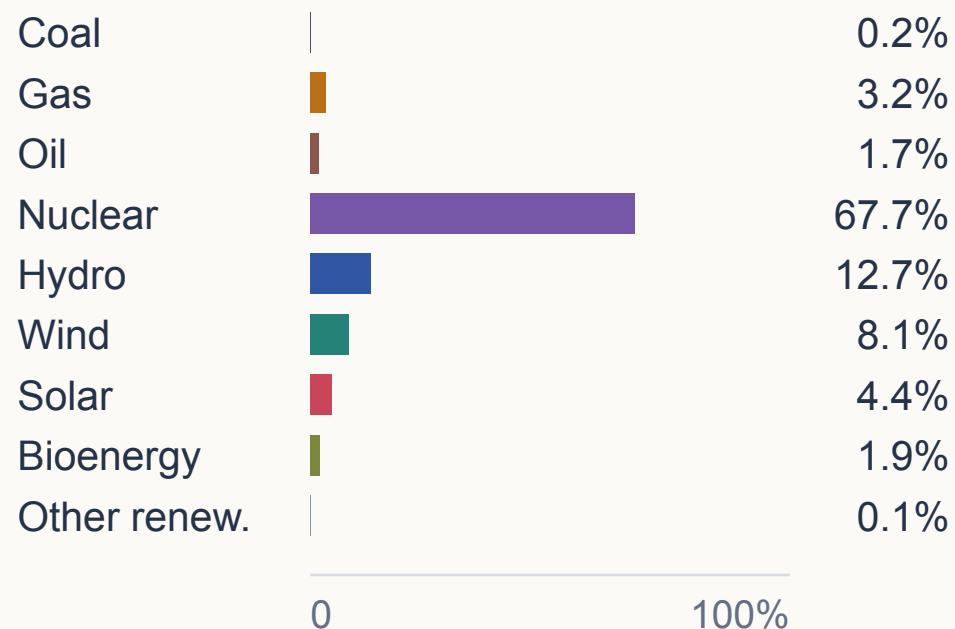
Small-source detail · 0–5%



Small slices offer little color area; a numeric label or detail view retains the value.

Radial bars give every source a fixed angular slot

Bar glyph · France, 2024

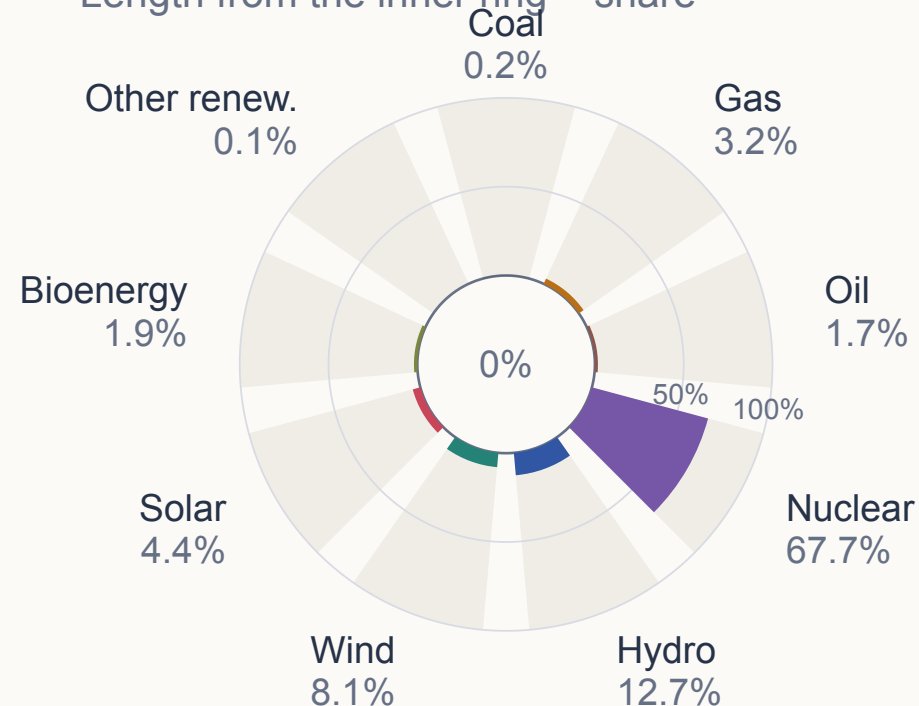


Same values, colors and 0–100% scale.

Short radial bars still need labels or detail. Read radial length, not sector area.

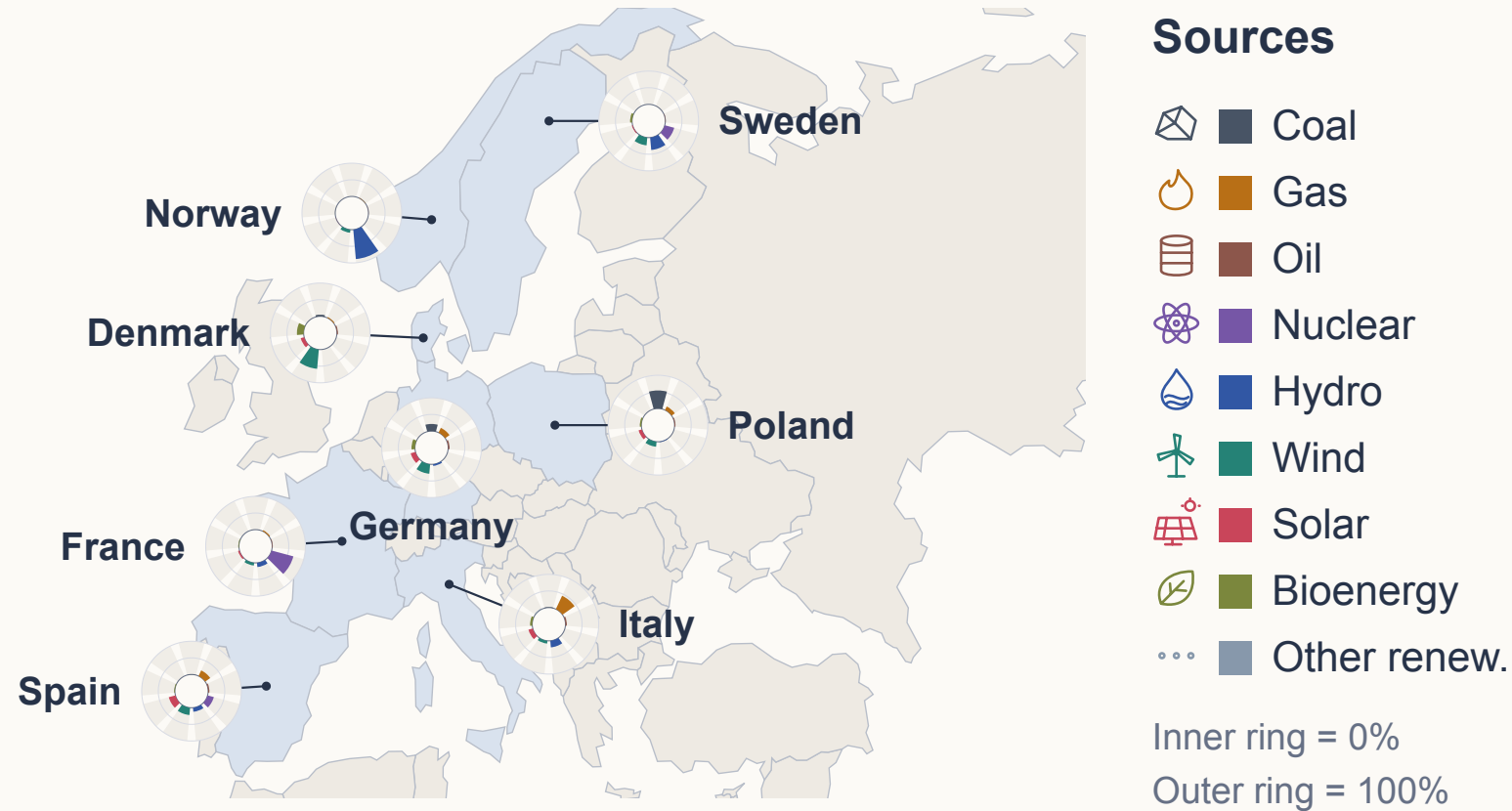
Radial bar glyph · same data

Length from the inner ring = share



Fixed slots replace narrow pie slices.

Radial bar glyphs compare electricity mixes across countries



2024 · share of national electricity generation · common 0–100% scale

Fixed source order · radial length = share · pale tracks = category slots

Critique explains a judgment; redesign acts on it

Stage	Principle	Concrete output
1 · Understanding	Infer knowledge, purpose and reading context from evidence.	An audience profile and a stated task.
2 · Task Fit	Describe style with the Wheel and judge the tradeoffs.	A reasoned target for the design.
3 · Encoding	Check expressiveness, effectiveness, separability and capacity.	A specific revision, followed by reassessment.

Visible evidence → reading consequence → design judgment → revision